TEACHER GUIDE

AQA GCSE Business (8132)

Break-even (financial calculations)



3.6 Finance

3.6.3 Financial terms and calculations: break-even

A revision lesson. Pupils run a car dealership in firms of four and revise break-even (financial calculations) on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Profit on the chart

Round 1

Your break-even chart.

Profit at that output? Minus if below zero.

Cars sold	**10 cars**
Total revenue at that output	**£220,000**
Fixed costs	**£4,250**
Variable costs at that output	**£210,000**

ANSWER

£5,750

£220,000 − £4,250 − £210,000 = £5,750

Why: Profit is the gap between the revenue line and the total cost line.

2 Lower break-even

Round 1

You want to lower your break-even point.

Which two changes would lower it?

- **A** Pay more for each car
- **B** Sell more cars
- **C** Pay higher rent
- **D** Raise the selling price
- **E** Cut the fixed costs

ANSWER

- ✓ **Raise the selling price**
- ✓ **Cut the fixed costs**

Why: More added per car or lower fixed costs; selling more does not move it.

3 Margin of safety

Round 2

Three charts: what you sell, and the break-even output.

Order by margin of safety, largest first?

- **A** Sell 20 cars, break-even 15
- **B** Sell 14 cars, break-even 6
- **C** Sell 9 cars, break-even 6

ANSWER

1. **Sell 14 cars, break-even 6**
2. **Sell 20 cars, break-even 15**
3. **Sell 9 cars, break-even 6**

Why: Sales minus break-even: 8, 5 and 3 cars. The most sales is not the safest.

Topic tasks

4 Profit on the deal

Round 2

A deal with a set-up cost.

Profit on the deal?

Livery and set-up, one-off	**£7,000**
Customer pays per car	**£10,750**
You pay per car	**£9,000**
They order	**9 cars**

ANSWER

£8,750

9 × (£10,750 − £9,000) − £7,000 = £8,750

Why: The first cars pay the set-up; the rest is profit: £8,750.

5 Less risky deal

Round 3

Both deals are for 10 cars. The taxi deal breaks even at 4 cars, the hire deal at 8.

Which deal is less risky, and why?

A The taxi deal

B The hire deal

ANSWER

✔ **The taxi deal**

Why: The taxi deal's margin of safety is 6 cars; the hire deal's only 2.

6 Profit at the sales

Round 3

The chart shows your costs and revenue at 5 cars.

Profit at the cars you expect to sell? Minus if below zero.

Fixed costs	**£13,250**
Total costs at 5 cars	**£50,750**
Total revenue at 5 cars	**£50,000**
Cars you expect to sell	**9 cars**

ANSWER

£9,250

(£50,750 − £13,250) ÷ 5 = £7,500 a car; £50,000 ÷ 5 = £10,000; 9 × £2,500 − £13,250 = £9,250

Why: Each car adds £2,500 above its own cost; the fixed costs come off once.

Final questions

Each pupil answers 6 final questions on break-even (financial calculations). They come from these 8 questions.

1 At the break-even level of output, what is true for a dealership?
✔ **Total revenue equals total costs**

2 A dealer's break-even output is 20 cars a month. In May it sells 24 cars. What does this mean?
✔ **It makes a profit in May**

3 A dealer's showroom rent rises. Nothing else changes. What happens to its break-even output?
✔ **It rises**

4 A valeting firm raises its price per valet. Nothing else changes. What happens to its break-even output?
✔ **It falls**

5 On a break-even chart, where is the break-even point?
✔ **Where the total revenue line crosses the total costs line**

6 A dealer's break-even output is 15 cars a month. In March it sells 11 cars. Which statement is correct?
✔ **It makes a loss in March**

7 A cheaper parts supplier cuts the variable cost of each service. Nothing else changes. What happens to break-even output?
✔ **It falls**

8 Why does a new dealer work out its break-even output before opening?
✔ **To know how many cars it must sell to avoid a loss**